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Dear Editors,

We submit the accompanying manuscript, “Which quantum algorithms suit photonic continuous-variable hardware? A comparative resource and noise-robustness study with a worked spectroscopic demonstration on  $\text{H}_2$ ,  $\text{HF}$ , and  $\text{I}_2$ ,” for consideration as a Paper in *New Journal of Physics*. The work falls within the journal’s broad scope of quantum information science, photonic quantum technologies, and computational molecular physics.

**Motivation.** Photonic continuous-variable (CV) architectures support deterministic, room-temperature generation of large cluster states and are a leading route to fault-tolerant quantum computing, where overhead is dominated by squeezing and by the consumption of non-Gaussian resource states. Different algorithms place very different demands on these resources, yet no unified comparison has quantified which algorithm classes are intrinsically robust on CV hardware and how much Gottesman–Kitaev–Preskill (GKP) error correction actually helps.

**Main results.** We port four representative algorithms—control-free phase estimation, randomized product formulas, quantum Kolmogorov–Arnold networks, and Regev factoring—to a common CV model and compare them under photon loss (exact Fock-space Kraus channel) and Gaussian displacement with GKP correction. We establish: (i) control-free phase estimation is the most noise-robust because it needs no controlled unitaries and, for quadratic Hamiltonians, no non-Gaussian gates, a property that survives for non-Gaussian Hamiltonians; (ii) GKP correction reduces residual error only above the finite-squeezing floor (proved), with the gain set by a noise- versus systematic-limited dichotomy whose empirical envelope constants we report with  $R^2 \geq 0.89$ ; (iii) the comb readout of Regev requires squeezing scaling as  $\sqrt{n}$  (proved); (iv) against a qubit surface-code baseline, the CV advantage is structural—Gaussian operations are free—rather than threshold-based.

**Worked physics example.** The manuscript includes a quantitative demonstration on three diatomic molecules ( $\text{H}_2$ ,  $\text{HF}$ ,  $\text{I}_2$ ) spanning a decade in vibrational quantum ( $\omega_e = 214\text{--}4401\text{ cm}^{-1}$ ) and an order of magnitude in anharmonicity. Using experimentally fixed Huber–Herzberg Morse parameters (no free parameters), CV control-free QPE recovers the populated vibrational eigenenergies of all three molecules to within  $0.55\text{ cm}^{-1}$ —below the standard  $1\text{ cm}^{-1}$  spectroscopic threshold; weakly anharmonic  $\text{I}_2$  reaches  $\sim 10^{-2}\text{ cm}^{-1}$ . Reaching the same accuracy with a discrete-variable QPE requires  $\sim 1.5 \times 10^{10}$  T-gates and  $\sim 16,000$  controlled time evolutions, while the CV control-free protocol needs zero controlled unitaries and zero non-Gaussian gates for the harmonic part. A Chebyshev-KAN classifier trained on the recovered eigenstates achieves 100% accuracy in assigning vibrational quantum numbers under realistic measurement noise.

**Methodological rigour.** The core GKP-correction surrogate is validated against an explicit Glancy–Knill error-correction Monte Carlo to within 0.1%, lifting a common methodological concern in CV-resource analyses. A two-dimensional noise phase diagram and Hamiltonian ensembles confirm the predictions are not artefacts of single operating points or single instances. We are explicit about which results are exact, which are validated model surrogates, and which are extrapolations.

**Fit to NJP.** New Journal of Physics has a long tradition of publishing methodological and computational advances at the intersection of quantum information and physical applications. We believe the manuscript fits this profile: it provides a unified methodology for assessing CV algorithm suitability, together with a quantitative physics application that yields concrete predictions experimentalists can test on near-term GKP-encoded photonic processors. All scripts and numerical results that reproduce every figure and table accompany the submission.

The manuscript has not been published and is not under consideration at any other journal; the authors declare no competing interests.

Sincerely,

Hikaru Wakaura  
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